

Lifting the exponent

Nyan

February 19, 2026

(AIME 2020)

Let n be the least positive integer for which $149^n - 2^n$ is divisible by $3^3 \cdot 5^5 \cdot 7^7$. Find the number of positive divisors of n .

Intuitively you just deal with the different prime powers separately.

Notation

$\nu_p(n)$ denotes the exponent of p in the prime factor of n . For example, $\nu_2 1434 = 1$.

LTE Statement

Let p be an odd prime and suppose x, y are ints with $p \mid x - y$, but not $p \mid x, y$. Then

$$\nu_p(x^n - y^n) = \nu_p(x - y) + \nu_p n$$

for all int n .

What happens when you take $x = 3, y = 1, n = 2$? The theorem breaks down actually. This is because all odd squares are $1 \pmod{8}$ and thus

$$8 \mid 3^2 - 1^2$$

while $\nu_2(3 - 1)$ is only 1.

Should tackle the case when $n \not\equiv 0 \pmod{p}$ separately. What happens when you take v_p of

$$\frac{x^n - y^n}{x - y} = x^{n-1} + \dots + y^{n-1}?$$

We're supposed to get 0- in other words, nonzero mod p .

However recall we assumed $p \mid x - y$ aka $y \equiv x \pmod{p}$. Thus the thing on the right just becomes

$$nx^{n-1}$$

which is obviously nonzero by definition.

So in other words, when not $p \mid n$ we have $v_p(x^n - y^n) = v_p(x - y)$.

We already tackled $p \nmid n$ earlier so now we can break down n as $p^{\nu_p n} n'$ for some integer n' not a multiple of p .

$$\nu_p(x^n - y^n) = \nu_p(x^{n/n'} - y^{n/n'}) = \nu_p(x^{p^{\nu_p n}} - y^{p^{\nu_p n}}).$$

So in other words we only care about prime powers n : want

$$\nu_p(x^{p^m} - y^{p^m}) = m + \nu_p(x - y)$$

It suffices to prove

Claim

$$\nu_p(x^p - y^p) = 1 + \nu_p(x - y).$$

Proof.

Much of the same. We do have to take some care to obtain exactly 1. We let $y = x + kp$ and calculate mod p^2 . In other words we want

$$x^{p-1} + \dots + y^{p-1} \equiv \text{smth} \cdot p \pmod{p^2}.$$

We can just get the other terms by cutting the $p^{\geq 2}$ terms automatically. Expand $y^n = (x + kp)^n$ by binomial theorem:

$$x^n + nx^{n-1}(kp) + \dots$$



Finish inductive step

Each of the terms is $(0 \leq a \leq p-1)$

$$x^{n-1-a}y^a \equiv x^{n-1-a}(x^a + ax^{a-1}kp) \pmod{p^2}$$

$$\text{Original} \equiv \sum_{a=0}^{p-1} (x^{p-1} + ax^{p-2}kp) = \sum x^{p-1} + kpx^{p-2} \sum a.$$

The latter thing is just zero so we end up with

$$px^{p-1} \pmod{p^2}$$

which is a multiple of p but not p^2 so we win.

Restate prob

Let n be the least positive integer for which $149^n - 2^n$ is divisible by $3^3 \cdot 5^5 \cdot 7^7$. Find the number of positive divisors of n .

- ◆ Need $\nu_3(149^n - 2^n) \geq 3$. By LTE this is $\nu_3(149 - 2) + \nu_3 n$. In other words we need 9 to divide n .
- ◆ Do the same thing for 7. Then $2 + \nu_7 n \geq 7$ so $7^5 \mid n$.
- ◆ When we do $p = 5$, notice that not $5 \mid 149 - 2$. It turns out that in order for

$$5 \mid 149^n - 2^n$$

we need $4 \mid n$. So if we let $n' = n/4$ then

$$5^5 \mid (149^4)^{n'} - (2^4)^{n'}$$

and by the exact same logic we get $5^4 \mid n'$.

To summarize, all the above forces $3^2 \cdot 7^5 \cdot 4 \cdot 5^4$ to divide n and that number is just the smallest n . The answer ends up being $3 \cdot 6 \cdot 3 \cdot 5 = \boxed{270}$.